



BIO 004 • HUMAN ANATOMY • SOLANO COMMUNITY COLLEGE

Module 2 Lab Structure List.

Bone • Skeleton • Joints

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How to use this list

This is a quick reference lab study guide. These are the lab structures and concepts to learn for the lab portion of the exam. Refer to the lab worksheets for the study questions that matter for the course. Treat this as the master list. Occasionally a lab sheet will not contain a structure seen here, so always cross reference against this list. Be prepared to identify structures on models, images, the cadaver, and slides from the microscope.

Some items are tagged with the module they are tested in. You meet the slide in Module 1 and are examined on it later.

Each tissue gets its own block. Tick the box by the name once you can identify it on the slide, write in its function and location, then work down the vertical list and tick each structure you can point to. Anything not in a list is context, not something you will be asked to identify.

READ THIS PART TWICE

This is an extensive list. It would be impossible to test on everything, but everything on this list is testable and you should prepare for it. On occasion I may add something to the list or remove something. Make sure you are in class, and have a person to connect with if you miss class, who can update you on any changes.

CHAPTER 1

Bone Tissue and the Long Bone

MICROSCOPIC AND GROSS

Osseous tissue on the model and the slide, then the whole long bone. Two questions on this sheet ask you to name a cartilage type, so answer them from the tissue chapter.

☐ Osseous tissue, on the bone tissue model

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Osteon | ☐ Osteocytes | ☐ Medullary cavity |
| ☐ Central (Haversian) canal | ☐ Inner circumferential lamellae | ☐ Periosteum, inner osteogenic layer |
| ☐ Volkmann's canal | ☐ Outer circumferential lamellae | ☐ Periosteum, outer fibrous layer |
| ☐ Canaliculi | ☐ Interstitial lamellae | ☐ Sharpey's fibers |
| ☐ Lacunae | ☐ Concentric lamellae | |

☐ Gross anatomy of a long bone

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Epiphysis | ☐ Periosteum, osteogenic layer | ☐ Compact bone |
| ☐ Metaphysis | ☐ Periosteum, fibrous layer | ☐ Spongy bone |
| ☐ Diaphysis | ☐ Articular cartilage | ☐ Collagen fibers between the lamellae |
| ☐ Medullary cavity | ☐ Epiphyseal line, in adults | ☐ Red marrow |
| ☐ Endosteum | ☐ Epiphyseal plate, in children | ☐ Yellow marrow |

Articular cartilage. Which type of cartilage is it? _____

Epiphyseal plate. Which type of cartilage is it? _____

CHAPTER 2

Axial Skeleton: The Skull

CRANIAL AND FACIAL BONES, SUTURES, FORAMINA

The largest single block of identifications in the course. Work it view by view, not bone by bone, because that is how it is tagged.

☐ Cranial and facial bones

STRUCTURES TO IDENTIFY

- | | | |
|------------------------------------|------------------------------------|-----------------------------------|
| ☐ Frontal | ☐ Sphenoid | ☐ Nasal |
| ☐ Occipital | ☐ Ethmoid | ☐ Lacrimal |
| ☐ Parietal | ☐ Maxilla | ☐ Mandible |
| ☐ Temporal | ☐ Zygomatic | |

☐ Sutures

STRUCTURES TO IDENTIFY

- | | |
|-----------------------------------|------------------------------------|
| ☐ Coronal | ☐ Squamosal |
| ☐ Sagittal | ☐ Lambdoid |

☐ Lateral skull

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Glabella | ☐ Zygomatic process of the temporal bone | ☐ Styloid process |
| ☐ Supraorbital foramen or notch | ☐ Temporal process of the zygomatic bone | ☐ Articular tubercle |
| ☐ Supraorbital margin | ☐ Mastoid process | ☐ Alveolar process |
| ☐ Zygomatic arch | ☐ External acoustic (auditory) meatus | ☐ Anterior nasal spine |
| | | ☐ Infraorbital foramen |

☐ Anterior skull

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Supraorbital foramen or notch | ☐ Middle nasal concha | ☐ Alveolar process |
| ☐ Supraorbital margin | ☐ Inferior nasal concha | ☐ Infraorbital foramen |
| ☐ Glabella | ☐ Middle nasal meatus | ☐ Nasion |
| ☐ Perpendicular plate of the ethmoid | ☐ Inferior nasal meatus | |
| ☐ Vomer | ☐ Anterior nasal spine | |

☐ Posterior skull

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Superior nuchal line | ☐ Vomer | ☐ Palatine process of the maxilla |
| ☐ Inferior nuchal line | ☐ Occipital condyles | ☐ Incisive fossa and foramen |
| ☐ External occipital protuberance | ☐ Palatine bone | ☐ Mastoid process |

☐ **Base of the skull, exterior****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Incisive foramen | ☐ Foramen lacerum | ☐ Mastoid process |
| ☐ Transverse palatine suture | ☐ Foramen rotundum | ☐ Articular tubercle |
| ☐ Median palatine suture | ☐ Carotid canal | ☐ Occipital condyles |
| ☐ Palatine bone | ☐ Jugular foramen | ☐ Styloid process |
| ☐ Vomer | ☐ Hypoglossal canal | ☐ Mandibular fossa |
| ☐ Medial pterygoid plate | ☐ Foramen magnum | ☐ Zygomatic arch |
| ☐ Lateral pterygoid plate | ☐ Inferior nuchal line | ☐ Inferior orbital fissure |
| ☐ Foramen ovale | ☐ Superior nuchal line | ☐ Choana |
| ☐ Foramen spinosum | ☐ External occipital protuberance | |

☐ **Base of the skull, interior****STRUCTURES TO IDENTIFY**

- | | | |
|--|--|---|
| ☐ Anterior cranial fossa | ☐ Lesser wing of the sphenoid | ☐ Hypoglossal canal |
| ☐ Middle cranial fossa | ☐ Greater wing of the sphenoid | ☐ Foramen lacerum |
| ☐ Posterior cranial fossa | ☐ Hypophyseal fossa (sella turcica) | ☐ Foramen magnum |
| ☐ Frontal crest | ☐ Dorsum sellae | ☐ Foramen spinosum |
| ☐ Frontal sinus | ☐ Tuberculum sellae | ☐ Foramen ovale |
| ☐ Cribriform plate of the ethmoid | ☐ Internal acoustic (auditory) meatus | ☐ Optic canal |
| ☐ Crista galli | ☐ Jugular foramen | ☐ Anterior clinoid processes |
| ☐ Olfactory foramina | ☐ Internal occipital protuberance | |

Which structure sits in the hypophyseal fossa? _____

☐ **Ethmoid bone****STRUCTURES TO IDENTIFY**

- | | |
|--|---|
| ☐ Crista galli | ☐ Cribriform plate |
| ☐ Perpendicular plate | ☐ Olfactory foramina |

☐ **Sphenoid bone****STRUCTURES TO IDENTIFY**

- | | | |
|--|--|---|
| ☐ Medial pterygoid plate | ☐ Dorsum sellae | ☐ Lesser wing |
| ☐ Lateral pterygoid plate | ☐ Tuberculum sellae | ☐ Superior orbital fissure |
| ☐ Hypophyseal fossa (sella turcica) | ☐ Greater wing | |

☐ **Orbit****STRUCTURES TO IDENTIFY**

- | | | |
|--|--|---|
| ☐ Sphenoid bone | ☐ Lacrimal canal | ☐ Superior orbital fissure |
| ☐ Lacrimal bone | ☐ Optic canal (foramen) | ☐ Inferior orbital fissure |

☐ **Mandible****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|--|
| ☐ Condylar process (condyle) | ☐ Mental foramen | ☐ Ramus |
| ☐ Coronoid process | ☐ Alveolar process | ☐ Mandibular foramen |
| ☐ Mandibular notch | ☐ Alveolar margin | ☐ Angle of the mandible |
| ☐ Mental protuberance | ☐ Body | |

☐ **Hyoid bone****STRUCTURES TO IDENTIFY**

- | | | |
|-------------------------------|---|---|
| ☐ Body | ☐ Greater cornu (greater horn) | ☐ Lesser cornu (lesser horn) |
|-------------------------------|---|---|

☐ **Nasal septum****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|--------------------------------|
| ☐ Septal cartilage | ☐ Perpendicular plate of the ethmoid | ☐ Vomer |
|---|---|--------------------------------|

☐ **Fetal skull****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Anterior fontanel | ☐ Right and left anterolateral fontanel | ☐ Right and left posterolateral fontanel |
| ☐ Posterior fontanel | | |

CHAPTER 3**Axial Skeleton: Spine and Thoracic Cage****VERTEBRAE, SACRUM, RIBS, STERNUM**

Identify the segment, then the region, then the individual vertebra. The regional features are what let you place a single loose vertebra.

☐ **Spinal segments****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Cervical spine, C1 to C7 | ☐ Lumbar spine, L1 to L5 | ☐ Coccyx, fused |
| ☐ Thoracic spine, T1 to T12 | ☐ Sacrum, S1 to S5 fused | |

☐ **Normal spinal curves**

Label each as primary or secondary. Know which are present at birth, and which developmental milestone produces each secondary curve, such as holding the head up, sitting, and walking.

STRUCTURES TO IDENTIFY

- | | |
|-----------------------------------|---|
| ☐ Cervical | ☐ Lumbar |
| ☐ Thoracic | ☐ Sacrococcygeal |

☐ **Abnormal spinal curves**

Define each as well as identify it.

STRUCTURES TO IDENTIFY

- | | | |
|------------------------------------|-----------------------------------|-----------------------------------|
| ☐ Scoliosis | ☐ Lordosis | ☐ Kyphosis |
|------------------------------------|-----------------------------------|-----------------------------------|

☐ **C1, the atlas****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|---|
| ☐ Anterior tubercle | ☐ Posterior arch | ☐ Transverse foramen |
| ☐ Posterior tubercle | ☐ Superior articular facet and process | ☐ Vertebral foramen |
| ☐ Facet for the dens | ☐ Inferior articular facet and process | |
| ☐ Anterior arch | ☐ Transverse process | |

☐ **C2, the axis****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|--|
| ☐ Dens | ☐ Body | ☐ Vertebral foramen |
| ☐ Superior articular facet and process | ☐ Transverse process | ☐ Spinous process |
| ☐ Inferior articular facet and process | ☐ Transverse foramen | ☐ Vertebral arch |

☐ **Typical cervical vertebra****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Vertebral foramen | ☐ Pedicle | ☐ Superior articular facet and process |
| ☐ Transverse process | ☐ Lamina | ☐ Inferior articular facet and process |
| ☐ Transverse foramen | ☐ Spinous process | |
| ☐ Body | ☐ Vertebral arch | |

☐ **Typical thoracic and lumbar vertebra****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|--|
| ☐ Body | ☐ Transverse process | ☐ Vertebral arch |
| ☐ Superior articular facet and process | ☐ Spinous process | ☐ Vertebral foramen |
| ☐ Inferior articular facet and process | ☐ Lamina | |
| ☐ Costal facet, thoracic only | ☐ Pedicle | |

☐ **Placing a vertebra on the articulated column**

Identify specific vertebrae on an articulated column, for example C7, T5, L1. Then work the other way: from the features alone, name the region. A transverse foramen means cervical. A costal facet means thoracic. A bifid spinous process means cervical. A spinous process pointing straight back means lumbar.

☐ **Rib****STRUCTURES TO IDENTIFY**

- | | | |
|-------------------------------|-------------------------------|--------------------------------|
| ☐ Head | ☐ Neck | ☐ Shaft |
|-------------------------------|-------------------------------|--------------------------------|

☐ **Sacrum and coccyx****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Sacral promontory | ☐ Posterior sacral foramina | ☐ Articular surface |
| ☐ Superior articular process and facet | ☐ Sacral hiatus | ☐ Base of the sacrum |
| ☐ Anterior sacral foramina | ☐ Median sacral crest | ☐ Apex of the sacrum |
| ☐ Coccyx | ☐ Lateral sacral crest | ☐ Sacral tuberosity |

☐ **Sternum**

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---------------------------------------|
| ☐ Manubrium | ☐ Xiphoid process | ☐ Costal notch |
| ☐ Sternal angle | ☐ Jugular notch | |
| ☐ Body | ☐ Clavicular notch | |

☐ **Intervertebral disc**

STRUCTURES TO IDENTIFY

- | | |
|---|---|
| ☐ Annulus fibrosus | ☐ Nucleus pulposus |
|---|---|

CHAPTER 4

Appendicular Skeleton: Pectoral Girdle and Upper Limb

CLAVICLE TO PHALANGES

Girdle first, then limb, then hand. The carpals have a memory tool built into the list.

☐ **Clavicle**

STRUCTURES TO IDENTIFY

- | | | |
|--------------------------------------|---------------------------------------|--|
| ☐ Sternal end | ☐ Acromial end | ☐ Conoid tubercle |
|--------------------------------------|---------------------------------------|--|

☐ **Scapula**

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ Acromion | ☐ Subscapular fossa | ☐ Superior angle |
| ☐ Coracoid process | ☐ Glenoid cavity | ☐ Suprascapular notch |
| ☐ Spine of the scapula | ☐ Supraglenoid tubercle | ☐ Axillary (lateral) border |
| ☐ Infraspinous fossa | ☐ Infraglenoid tubercle | ☐ Vertebral (medial) border |
| ☐ Supraspinous fossa | ☐ Superior border | ☐ Inferior angle |

☐ **Humerus**

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Head | ☐ Deltoid tuberosity | ☐ Trochlea |
| ☐ Anatomical neck | ☐ Radial groove | ☐ Coronoid fossa |
| ☐ Surgical neck | ☐ Lateral epicondyle | ☐ Capitulum |
| ☐ Greater tubercle | ☐ Lateral supracondylar ridge | ☐ Radial fossa |
| ☐ Lesser tubercle | ☐ Medial epicondyle | ☐ Olecranon fossa |
| ☐ Intertubercular sulcus (bicipital groove) | ☐ Medial supracondylar ridge | |

☐ **Radius**

STRUCTURES TO IDENTIFY

- | | | |
|-------------------------------|--|--------------------------------------|
| ☐ Head | ☐ Styloid process | ☐ Ulnar notch |
| ☐ Neck | ☐ Radial tuberosity | |

☐ **Ulna****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|--|
| ☐ Head | ☐ Radial notch | ☐ Olecranon process |
| ☐ Styloid process | ☐ Coronoid process | |
| ☐ Ulnar tuberosity | ☐ Trochlear notch | |

☐ **Carpals, on an articulated wrist**

Memory tool: So Long To Pinky, proximal row lateral to medial. Here Comes The Thumb, distal row medial to lateral.

STRUCTURES TO IDENTIFY

- | | | |
|--|-----------------------------------|------------------------------------|
| ☐ Scaphoid | ☐ Pisiform | ☐ Trapezoid |
| ☐ Lunate | ☐ Hamate | ☐ Trapezium |
| ☐ Triquetrum (triangular) | ☐ Capitate | |

☐ **Metacarpals and phalanges**

The first digit has no middle phalanx.

STRUCTURES TO IDENTIFY

- | | |
|--|---|
| ☐ Metacarpals 1 to 5, lateral to medial | ☐ Middle phalanges |
| ☐ Proximal phalanges | ☐ Distal phalanges |

CHAPTER 5**Appendicular Skeleton: Pelvic Girdle and Lower Limb****COXAL BONE TO PHALANGES**

The coxal bone is three fused bones, so learn it as three regions that meet at the acetabulum.

☐ **Coxal bone, the ilium region****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Iliac crest | ☐ Lesser sciatic notch | ☐ Posterior superior iliac spine (PSIS) |
| ☐ Auricular surface | ☐ Ala | ☐ Posterior inferior iliac spine (PIIS) |
| ☐ Iliac fossa | ☐ Anterior superior iliac spine (ASIS) | |
| ☐ Greater sciatic notch | ☐ Anterior inferior iliac spine (AIIS) | |

☐ **Ischium****STRUCTURES TO IDENTIFY**

- | | |
|---|---|
| ☐ Ischial spine | ☐ Lesser sciatic notch |
| ☐ Ischial tuberosity | ☐ Ramus of the ischium |

☐ **Pubis****STRUCTURES TO IDENTIFY**

- | | | |
|---|---|---|
| ☐ Pubic body | ☐ Inferior ramus | ☐ Pubic tubercle |
| ☐ Superior ramus | ☐ Pubic crest | |

☐ **Landmarks and joints of the pelvis****STRUCTURES TO IDENTIFY**

- | | |
|--|---|
| ☐ Acetabulum | ☐ Sacroiliac joint |
| ☐ Obturator foramen | ☐ Pubic symphysis |

☐ **Femur****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|--|
| ☐ Head | ☐ Gluteal tuberosity | ☐ Medial epicondyle |
| ☐ Fovea capitis | ☐ Linea aspera, lateral lip | ☐ Lateral condyle |
| ☐ Neck | ☐ Linea aspera, medial lip | ☐ Medial condyle |
| ☐ Greater trochanter | ☐ Lateral supracondylar line | ☐ Patellar surface |
| ☐ Lesser trochanter | ☐ Medial supracondylar line | ☐ Intercondylar fossa |
| ☐ Intertrochanteric line | ☐ Adductor tubercle | |
| ☐ Intertrochanteric crest | ☐ Lateral epicondyle | |

☐ **Patella**☐ **Tibia****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Lateral condyle | ☐ Tibial tuberosity | ☐ Medial malleolus |
| ☐ Medial condyle | ☐ Fibular notch | |
| ☐ Intercondylar eminence | ☐ Anterior crest (border) | |

☐ **Fibula****STRUCTURES TO IDENTIFY**

- | | |
|---|--|
| ☐ Head of the fibula | ☐ Lateral malleolus |
|---|--|

☐ **Tarsals**

Memory device: the three C's surround the N and T.

STRUCTURES TO IDENTIFY

- | | | |
|------------------------------------|---|---------------------------------|
| ☐ Talus | ☐ Medial cuneiform | ☐ Cuboid |
| ☐ Calcaneus | ☐ Intermediate cuneiform | |
| ☐ Navicular | ☐ Lateral cuneiform | |

☐ **Metatarsals and phalanges**

The first digit, the hallux, has only a proximal and a distal phalanx. Toes 2 to 5 have all three.

STRUCTURES TO IDENTIFY

- | | |
|--|---|
| ☐ Metatarsals I to V, medial to lateral | ☐ Middle phalanges |
| ☐ Proximal phalanges | ☐ Distal phalanges |

CHAPTER 6**Joints****CLASSIFICATION, NAMED JOINTS, THE KNEE**

For any tagged joint you may be asked four things. Practise answering all four every time, not just the name.

For any joint you are given, be ready to

1. Classify it as fibrous, cartilaginous, or synovial.
2. Name the type of synovial joint, for example hinge, ball and socket, saddle.
3. Classify it by movement: synarthrosis (immovable), amphiarthrosis (slightly movable), diarthrosis (freely movable).
4. Name the specific movements that occur there, if any. Flexion, extension, internal rotation, and so on.

☐ **Named joints, know these when tagged**

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Atlantoaxial | ☐ Sternoclavicular | ☐ Tibiofibular |
| ☐ Atlanto-occipital | ☐ Manubriosternal | ☐ Temporomandibular |
| ☐ Glenohumeral | ☐ Radioulnar | ☐ Sacroiliac |
| ☐ Acromioclavicular | ☐ Talocrural | ☐ Pubic symphysis |

☐ **A typical synovial joint**

STRUCTURES TO IDENTIFY

- | | |
|--|---|
| ☐ Articular cartilage | ☐ Joint cavity |
| ☐ Synovial membrane | ☐ Synovial fluid |

☐ **The knee joint**

STRUCTURES TO IDENTIFY

- | | | |
|----------------------------------|--|--|
| ☐ Femur | ☐ Patellar ligament | ☐ Medial meniscus |
| ☐ Tibia | ☐ Quadriceps tendon | ☐ Lateral meniscus |
| ☐ Fibula | ☐ Medial (tibial) collateral ligament | ☐ Anterior cruciate ligament |
| ☐ Patella | ☐ Lateral (fibular) collateral ligament | ☐ Posterior cruciate ligament |